

Task 5: Game Theory Model Proof

Paper: Feng, Liu, Zhang, and Srinivasan, "Sustainability and Competition on Amazon."

Task: evaluate the game theory model. Determine whether it can simultaneously explain the four empirical findings below. If not, propose modifications that would allow it to account for all four simultaneously.

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1. Model Mechanism Summary

The badge in this model is not a signal. There is no private information in the game: sustainability levels f_A and f_B , all preference parameters, and the threshold I_0 are common knowledge (Figure 1). The badge operates through two direct channels in a Singh and Vives (1984) linear quadratic duopoly. First, it is a utility shifter: for the green segment of size k , badged product j gains an additive term $\rho_i \cdot G_j$ in marginal utility (Eq. 1). Second, and doing most of the work, it is a participation gate: green consumers are assumed to buy only badged products and to abstain entirely from unbadged ones (assumption stated after Eqs. 4 and 5). Non green consumers (size $1 - k$) ignore the badge entirely (Eq. 6).

Badging therefore expands or reallocates the effective market: a badged product sells to both segments, an unbadged product sells only to non green consumers. The game is solved by backward induction for subgame perfect equilibrium: the marketplace picks I_0 from the coarse set $\{0, 0.5, 1\}$ (Section 3.3), sellers Bertrand compete in prices (Lemmas 1 to 3), and consumers purchase. All comparative statics on prices and demand flow from the $k \cdot \rho_0$ terms the badge injects into the demand intercepts. The participation gate is imposed, not derived from primitives. It is the engine of every demand expansion result below.

2. Finding by Finding Derivation

Finding 1: badge increases demand for badged sellers. Status: derived, conditional.

Situation 2 versus 1 (one product badged versus none): Eq. 24 gives $D_A(2) - D_A(1)$ proportional to $(2 - z^2) \alpha_{A0} > z \alpha_{B0} + (1 - z^2) \rho_0$, positive if and only if condition (25) holds. The paper's own decomposition is honest here: the influx of green consumers must outweigh the non green consumers lost to the equilibrium price increase. The unbadged product B gains demand unconditionally in situation 2 (Eq. 26, always positive) through a relative price effect. For situation 3 versus 1, the analogous conditions are Eqs. 33 and 34, summarized in Proposition 2. So Finding 1 is a genuine equilibrium result, but only inside the baseline utility dominance region.

Finding 2: badge increases price. Status: derived, unconditional.

Comparing Lemmas 1 and 2, Eq. 23 gives both $p_A^*(2) - p_A^*(1)$ and the analogous term for seller B strictly positive. The badged seller's price rises because its demand intercept rises with the green segment premium. The unbadged seller's price rises purely through strategic complementarity, since its best response is increasing in the rival's price. Findings 1 and 2 are jointly coherent, not smuggled in: both are consequences of the same intercept shift, and condition (25) correctly prices in the tension between them. The caveat inherited from Section 1 stands: the intercept shift rests on the assumed participation gate.

Finding 3: platform prefers selective badging. Status: partially derived, sufficient conditions only, no modeled trade off.

Proposition 3 states that situation 2 dominates both 1 and 3 when three conditions hold jointly: condition (25), the band in Eq. 38 which requires α_{A0} minus α_{B0} to sit in a strictly negative interval, and 0.67 less than z less than 0.89 . Two structural weaknesses stand out. First, footnote 4 concedes the platform's aggregate objective $r \cdot (p_A D_A + p_B D_B)$ is never compared across the three thresholds. The paper requires pointwise dominance of every price and every demand term simultaneously, a valid sufficient condition but not an actual profit maximization. Second, there

is no cost, congestion, or dilution externality attached to badging anywhere in the model. Nothing structural penalizes badging everyone. The interior optimum at I_0 equal to 0.5 exists only because the chosen parameter band happens to make situation 2 dominate pointwise. The trade off Finding 3 requires is assumed via parameter selection, not modeled.

Finding 4: badge decreases market concentration (HHI). Status: confirmed conditionally false inside part of Proposition 3's own band.

This section was re run with a full symbolic derivation, not taken on the paper's stated closed forms. One correction surfaced in the process: the badge premium term in the transcribed Lemma 2 formulas is written as a function of z (substitutability). Deriving the model from the underlying utility functions and checking it against the paper's own Situation 1 result shows the term must instead be a function of k (green segment size). The z reading fails to reduce correctly to Situation 1; the k reading does. This is most likely an OCR transcription error carried through both prior analyses of this paper, not an error in the paper itself. It matters because k , not z , is the variable that actually governs the HHI result below.

Using Proposition 3's own worked numerical example ($z = 0.8$, $\alpha_{B0} = 2$, $\rho_0 = 0.5$, α_{A0} in the band 1.56 to 1.90), I computed MS_A under Situations 1 and 2 directly from the derived demand functions and checked the sign of the gap to 0.5 across a range of k , the one free parameter Proposition 3 never fixes.

α_{A0}	$k = 0.1$	$k = 0.3$	$k = 0.5$	$k = 0.7$	$k = 0.9$
1.56	falls	falls	falls	falls	rises
1.65	falls	falls	falls	rises	rises
1.75	falls	falls	falls (barely)	rises	rises
1.90	falls	rises	rises	rises	rises

The result is conditional, not binary. HHI falls only when the green segment is a minority of the market, roughly k below 0.5 to 0.7, tightening further as α_{A0} rises toward the top of the band. Once green consumers are a majority, badging the weaker seller A pushes its market share past 0.5 by more than its starting deficit, and concentration rises instead of falling. This is the U shaped HHI curve working against the badge once the correction overshoots the balance point.

3. Consistency Verdict

The model, as written, cannot jointly generate all four findings under one internally consistent parameterization for every value of k . Findings 1, 2, and 3 hold throughout Proposition 3's band regardless of k , since none of those conditions involve k at all. Finding 4 holds only on a sub region of that same band, bounded by k . The paper never states or derives this restriction.

This is a sharper conclusion than either prior pass on this paper reached. The first pass on this paper flagged the tension between Proposition 3 and Proposition 4 as an open gap, unverified in either direction. An independent AI pass (Deepseek) reproduced the same tension in its own algebra, then asserted a resolution without deriving it, which is not a verified claim. The check run here closes the gap directly: Finding 4 is not simply unproven, it is conditionally false, and the condition is exactly identifiable (k above roughly 0.5 to 0.7, depending on α_{A0}).

Practical read: the paper's empirical badge proportion is 35.39 percent, a product level statistic, not the same object as k , the green consumer population share. It is plausible the real world k sits in the low range where the model's HHI prediction holds, which would explain why the empirical result (Table 7) shows HHI falling. But this is a coincidence of where the real data happens to sit, not a property the model guarantees. Proposition 3 and Proposition 4 are compatible only on a sub region of k that the paper never names.

4. Proposed Modifications, Ordered by Cost

Modification 1, lowest cost: state the k dependent boundary explicitly. Since the check above already identifies where the sign flips, the paper's Proposition 4 should be restated with an explicit upper bound on k , derived algebraically rather than left to a single numerical illustration with an unstated k .

Modification 2: replace pointwise dominance in Finding 3 with an actual Stage 1 objective. Compute platform profit $r \cdot (p_A D_A + p_B D_B)$ in each situation and take the argmax over I_0 . This upgrades Finding 3 from a sufficient condition to an actual optimum and may narrow or shift the region where the Modification 1 boundary also holds.

Modification 3, highest cost: extend to N heterogeneous sellers with a badge dilution externality, so the interior optimum for I_0 and the HHI recomposition both emerge from modeled economics rather than a hand picked parameter set. This also resolves the Eq. 41 HHI floor issue, since the $1/N$ floor only becomes meaningful for N of 4 or more.

5. Secondary Gaps Worth Noting, Not Central to the Verdict

Demographic heterogeneity: the model treats the green segment as one uniform k with constant ρ_i . The empirical result shows older males respond most, with women and younger consumers showing insignificant effects (Table 3). The model has no structure to produce this.

Seller strategic response: sustainability level f_A and marketing quality are exogenous in the model. Model 2 having a smaller ATT than Model 1 once images and descriptions are controlled for indicates part of the badge effect is mediated through seller choices the game never models.

6. Bottom Line

Accept Findings 1 and 2 as derived. Treat Finding 3 as a sufficient condition result pending an aggregate profit comparison. Finding 4 is not an open question anymore: it is confirmed to hold only for k below a specific, identifiable threshold inside Proposition 3's own band, and to fail above it. The one page fix is to state that threshold. The paper's four findings are jointly consistent only if the real world green segment share is smaller than the point where the model's own algebra flips sign, a condition the paper's empirical badge proportion (35.39 percent) is consistent with, but never confirms.